



Signals and Systems

Discrete Time Fourier Series

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Example 9: $x[n] = \text{Sin}(\frac{\pi n}{2})u[n]$

$$x[n] = \left(\frac{e^{j(\pi n/2)} - e^{-j(\pi n/2)}}{2j} \right) u[n]$$

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \left(\frac{e^{j(\pi n/2)} - e^{-j(\pi n/2)}}{2j} \right) u[n] e^{-j\omega n} \\ &= \frac{1}{2j} \sum_{n=0}^{\infty} (e^{j(\pi n/2)} e^{-j\omega n} - e^{-j(\pi n/2)} e^{-j\omega n}) \end{aligned}$$

$$\begin{aligned}
 X(\omega) &= \frac{1}{2j} \sum_{n=0}^{\infty} (e^{j(\pi/2)} e^{-j\omega})^n - (e^{-j(\pi/2)} e^{-j\omega})^n \\
 &= \frac{1}{2j} \left(\frac{1}{1 - e^{j(\pi/2)} e^{-j\omega}} - \frac{1}{1 - e^{-j(\pi/2)} e^{-j\omega}} \right)
 \end{aligned}$$

Example 10: $x[n] = \cos\left(\frac{\pi n}{2}\right) u[n]$

$$x[n] = \left(\frac{e^{j(\pi n/2)} + e^{-j(\pi n/2)}}{2} \right) u[n]$$

$$\begin{aligned}
 X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \left(\frac{e^{j(\pi n/2)} + e^{-j(\pi n/2)}}{2} \right) u[n] e^{-j\omega n} \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} (e^{j(\pi n/2)} e^{-j\omega n} + e^{-j(\pi n/2)} e^{-j\omega n})
 \end{aligned}$$

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 X(\omega) &= \frac{1}{2} \sum_{n=0}^{\infty} (e^{j(\pi/2)} e^{-j\omega})^n + (e^{-j(\pi/2)} e^{-j\omega})^n \\
 &= \frac{1}{2} \left(\frac{1}{1 - e^{j(\pi/2)} e^{-j\omega}} + \frac{1}{1 - e^{-j(\pi/2)} e^{-j\omega}} \right)
 \end{aligned}$$

$$\rightarrow F[\sin(\omega_0 n)u(n)] = \frac{1}{2j} \left(\frac{1}{1 - e^{j\omega_0} e^{-j\omega}} - \frac{1}{1 - e^{-j\omega_0} e^{-j\omega}} \right)$$

$$\rightarrow F[\cos(\omega_0 n)u(n)] = \frac{1}{2} \left(\frac{1}{1 - e^{j\omega_0} e^{-j\omega}} + \frac{1}{1 - e^{-j\omega_0} e^{-j\omega}} \right)$$

Properties of Discrete Time Fourier Transform

Notations to be used

$$F[x(n)] = X(e^{j\omega}) = X(\omega)$$

$$F^{-1}[X(\omega)] = x(n)$$

$$x(n) \xleftrightarrow{F} X(\omega)$$

1. Periodicity Property of DTFT

❖ Fourier transform $X(\omega)$ of Discrete time is periodic with period 2π .

$$X(\omega + 2\pi) = X(\omega)$$

2. Linearity Property

If $x_1(n) \xleftrightarrow{F} X_1(\omega)$ and $x_2(n) \xleftrightarrow{F} X_2(\omega)$

then

$$a_1 x_1(n) + a_2 x_2(n) \xleftrightarrow{F} a_1 X_1(\omega) + a_2 X_2(\omega)$$

Proof:

$$\begin{aligned} F[a_1 x_1(n) + a_2 x_2(n)] &= \sum_{n=-\infty}^{\infty} [a_1 x_1(n) + a_2 x_2(n)] e^{-j\omega n} \\ &= a_1 \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} + a_2 \sum_{n=-\infty}^{\infty} x_2(n) e^{-j\omega n} \\ &= a_1 X_1(\omega) + a_2 X_2(\omega) \end{aligned}$$

3. Time shifting Property

If $x(n) \xleftrightarrow{F} X(\omega)$

then

$$x(n - n_0) \xleftrightarrow{F} e^{-j\omega n_0} X(\omega)$$

Proof:

$$\begin{aligned} F[x(n - n_0)] &= \sum_{n=-\infty}^{\infty} x(n - n_0) e^{-j\omega n} && \text{put} \\ & && n - n_0 = p \\ &= \sum_{p=-\infty}^{\infty} x(p) e^{-j\omega(p+n_0)} \\ &= e^{-j\omega n_0} \sum_{p=-\infty}^{\infty} x(p) e^{-j\omega p} \\ &= e^{-j\omega n_0} \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = e^{-j\omega n_0} X(\omega) \end{aligned}$$

4. Frequency shifting Property

If $x(n) \xleftrightarrow{F} X(\omega)$

then

$$e^{j\omega_0 n} x(n) \xleftrightarrow{F} X(\omega - \omega_0)$$

Proof:

$$\begin{aligned} F[e^{j\omega_0 n} x(n)] &= \sum_{n=-\infty}^{\infty} e^{j\omega_0 n} x(n) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega - \omega_0)n} \\ &= X(\omega - \omega_0) \end{aligned}$$

5. Time Reversal Property

If $x(n) \xleftrightarrow{F} X(\omega)$

then

$$x(-n) \xleftrightarrow{F} X(-\omega)$$

Proof:

$$F[x(-n)] = \sum_{n=-\infty}^{\infty} x(-n)e^{-j\omega n} \quad \begin{array}{l} \text{put} \\ -n = p \end{array}$$

$$= \sum_{p=-\infty}^{-\infty} x(p)e^{j\omega p} = \sum_{p=-\infty}^{\infty} x(p)e^{-j(-\omega)p}$$

$$= X(-\omega)$$

6. Differencing Property

If $x(n) \xleftrightarrow{F} X(\omega)$

then

$$x(n) - x(n-1) \xleftrightarrow{F} (1 - e^{-j\omega})X(\omega)$$

Proof:
$$\begin{aligned} F[x(n) - x(n-1)] &= F[x(n)] - F[x(n-1)] \\ &= X(\omega) - e^{-j\omega}X(\omega) \quad (\text{Time shifting Property}) \\ &= (1 - e^{-j\omega})X(\omega) \end{aligned}$$

7. Differentiation in frequency domain

If $x(n) \xleftrightarrow{F} X(\omega)$

then

$$nx(n) \xleftrightarrow{F} j \frac{d}{d\omega} (X(\omega))$$

Proof:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$\frac{d}{d\omega} (X(\omega)) = \frac{d}{d\omega} \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right) = \sum_{n=-\infty}^{\infty} x(n) \frac{d}{d\omega} (e^{-j\omega n})$$

$$\frac{d}{d\omega} (X(\omega)) = \sum_{n=-\infty}^{\infty} x(n) (-jn) e^{-j\omega n} = -j \sum_{n=-\infty}^{\infty} nx(n) e^{-j\omega n} = -jF[nx(n)]$$

So $F[nx(n)] = j \frac{d}{d\omega} X(\omega)$

8. Conjugate Property (for complex function $x(n)$)

If
$$x(n) \xleftrightarrow{F} X(\omega)$$

then

$$x^*(n) \xleftrightarrow{F} X^*(-\omega)$$

Proof:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

$$X^*(\omega) = \left[\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \right]^* = \sum_{n=-\infty}^{\infty} x^*(n)e^{j\omega n}$$

$$X^*(-\omega) = \sum_{n=-\infty}^{\infty} x^*(n)e^{-j(\omega)n} = F[x^*(n)]$$

So
$$x^*(n) \xleftrightarrow{F} X^*(-\omega)$$

Que. Find the Fourier transform of following signals using property.

$$(i) \quad x[n] = a^{n-2}u[n-2] \quad |a| < 1$$

$$(viii) \quad x[n] = n \cos\left(\frac{\pi n}{2}\right)u[n]$$

$$(ii) \quad x[n] = a^{|n-2|} \quad |a| < 1$$

$$(ix) \quad x[n] = a^n \sin\left(\frac{\pi n}{2}\right)u[n]$$

$$(iii) \quad x[n] = a^{-n}u[-n] \quad |a| < 1$$

$$(iv) \quad x[n] = na^n u[n] \quad |a| < 1$$

$$(v) \quad x[n] = e^{j5n}a^n u[n] \quad |a| < 1$$

$$(vi) \quad x[n] = (n-2)a^{|n-2|} \quad |a| < 1$$

$$(vii) \quad x[n] = n \sin\left(\frac{\pi n}{2}\right)u[n]$$

$$(i) \quad x[n] = a^{n-2} u[n-2] \quad |a| < 1$$

As already know.

$$F[a^n u(n)] = \frac{1}{1 - ae^{-j\omega}}$$

By using time shifting prop. $x(n - n_0) \xleftrightarrow{F} e^{-j\omega n_0} X(\omega)$

$$F[a^{n-2} u(n-2)] = \frac{e^{-j2\omega}}{1 - ae^{-j\omega}}$$

$$(ii) \quad x[n] = a^{|n-2|} \quad |a| < 1$$

As already know.

$$F[a^{|n|}] = \frac{1 - a^2}{1 - 2a \cos \omega + a^2}$$

By using time shifting prop. $F[a^{|n-2|}] = \frac{e^{-2j\omega} (1 - a^2)}{1 - 2a \cos \omega + a^2}$

$$\text{(iii)} \quad x[n] = a^{-n}u[-n] \quad |a| < 1$$

$$x[n] = x_1(-n)$$

$$F[a^n u(n)] = \frac{1}{1 - ae^{-j\omega}}$$

where

$$x_1(n) = a^n u(n)$$

By using time reversal property. $x(-n) \xleftrightarrow{F} X(-\omega)$

$$F[a^{-n}u(-n)] = F[x_1(-n)] = \frac{1}{1 - ae^{j\omega}}$$

$$\text{(iv)} \quad x[n] = na^n u[n] \quad |a| < 1$$

differentiation

$$F[na^n u(n)] = j \frac{d}{d\omega} \left(\frac{1}{1 - ae^{-j\omega}} \right)$$

$$nx(n) \xleftrightarrow{F} j \frac{d}{d\omega} (X(\omega))$$

$$F[na^n u(n)] = j \frac{d}{d\omega} \left(\frac{1}{1 - ae^{-j\omega}} \right) = \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2}$$

$$(v) \quad x[n] = e^{j5n} a^n u[n] \quad |a| < 1$$

As already know.

$$F[a^n u(n)] = \frac{1}{1 - ae^{-j\omega}}$$

By using frequency shifting prop. $e^{j\omega_0 n} x(n) \xleftrightarrow{F} X(\omega - \omega_0)$

$$F[e^{j5n} a^n u(n)] = \frac{1}{1 - ae^{-j(\omega - 5)}}$$

(vi) $x[n] = (n-2)a^{|n-2|} \quad |a| < 1$

As already know.

$$F[a^{|n|}] = \frac{1-a^2}{1-2a \cos \omega + a^2}$$

let $x_1(n) = na^{|n|}$

$$X_1(\omega) = F[na^{|n|}] = j \frac{d}{d\omega} \left(\frac{1-a^2}{1-2a \cos \omega + a^2} \right)$$

So $x[n] = x_1(n-2) = (n-2)a^{|n-2|}$

So $X(\omega) = F[(n-2)a^{|n-2|}] = e^{-j2\omega} X_1(\omega)$

On solving you will get the solution X(w)

$$\text{(vii)} \quad x[n] = n \sin\left(\frac{\pi n}{2}\right) u[n]$$

As already know.

$$F[\sin(\omega_0 n) u(n)] = \frac{1}{2j} \left(\frac{1}{1 - e^{j\omega_0} e^{-j\omega}} - \frac{1}{1 - e^{-j\omega_0} e^{-j\omega}} \right)$$

$$F\left[\sin\left(\frac{\pi n}{2}\right) u(n)\right] = \frac{1}{2j} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} - \frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right)$$

$$F\left[n \sin\left(\frac{\pi n}{2}\right) u(n)\right] = j \frac{d}{d\omega} \left[\frac{1}{2j} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} - \frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right) \right]$$

$$F\left[n \sin\left(\frac{\pi n}{2}\right) u(n)\right] = \frac{1}{2} \frac{d}{d\omega} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} \right) - \frac{1}{2} \frac{d}{d\omega} \left(\frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right)$$

On Solving you will get your answer.

$$\text{(viii)} \quad x[n] = n \cos\left(\frac{\pi n}{2}\right) u[n]$$

As already know.

$$F[\cos(\omega_0 n) u(n)] = \frac{1}{2} \left(\frac{1}{1 - e^{j\omega_0} e^{-j\omega}} + \frac{1}{1 - e^{-j\omega_0} e^{-j\omega}} \right)$$

$$F\left[\cos\left(\frac{\pi n}{2}\right) u(n)\right] = \frac{1}{2} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} + \frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right)$$

$$F\left[n \cos\left(\frac{\pi n}{2}\right) u(n)\right] = j \frac{d}{d\omega} \left[\frac{1}{2} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} + \frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right) \right]$$

$$F\left[n \cos\left(\frac{\pi n}{2}\right) u(n)\right] = \frac{j}{2} \frac{d}{d\omega} \left(\frac{1}{1 - e^{j\pi/2} e^{-j\omega}} \right) + \frac{j}{2} \frac{d}{d\omega} \left(\frac{1}{1 - e^{-j\pi/2} e^{-j\omega}} \right)$$

On Solving you will get your answer.